# Reduction #3: Max-of-$K$ Correlated Gaussian Fields and the Extreme-Value Origin of Tempotron Fragmentation

Tempotron learning-surface program

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**Abstract**

The tempotron's binary decision is the sign of its maximum membrane potential over time, which reduces in the large-$N$ limit to the maximum of $K$ correlated standard Gaussian fields. Extreme-value theory (EVT) then yields a *leading-order, annealed* formula (prefactor undetermined) for the expected number of disconnected solution clusters as a function of load $\alpha$ and field count $K$:

$$N_{\text{clusters}}(\alpha, K) \;\approx\; \exp\bigl[N\, S_{\text{cl}}(\alpha, K)\bigr] \;=\; (\ln K)^{N/2} \cdot 2^{-N\alpha},$$

where the equality holds to leading order in $\ln K$ (an undetermined $O(1)$ prefactor is absorbed in the subleading $\frac{N}{2} \ln c$ term; see §3.3). Setting $N_{\text{clusters}} = 1$ recovers the capacity lift $\alpha_c(K) = (\ln \ln K)/(2 \ln 2)$, and the inter-cluster overlap $q_0 \sim \alpha / \ln K \to 0$ quantifies the fragmentation of the solution manifold. The core result—lift and shattering are one EVT effect counted two ways—is verified at three independent points: the Gumbel scale, the $K = 1$ perceptron limit, and the $S_{\text{cl}} = 0$ capacity crossing. An independent codex cross-check [9] shows that the commonly-cited additive offset $\alpha_0 \approx 2.58$ is a small-$K$ fit artifact: the first-moment calculation gives a third iterated-log subleading term, not a constant. Status: **PARTIAL**—leading scalings solid; $O(1)$ constants and quenched cluster complexity open.

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*Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.*

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# 1 Introduction

The tempotron [8] is a leaky integrate-and-fire neuron trained to fire (or remain silent) in response to an $N$-afferent spike-timing pattern. Its membrane potential at time $t$ is

$$V(t;\mathbf{w}) \;=\; \sum_{i=1}^{N} w_i\, K(t-s_i), \tag{1}$$

where $w_i$ are synaptic weights and $K$ is a double-exponential kernel. The neuron fires if $\max_t V(t;\mathbf{w})$ exceeds a fixed threshold $\vartheta$. Because the kernel is shared across synapses, the time-dependent maximum depends on the same weight vector for all $t$; this shared-weight structure is what makes the problem interesting and hard.

Rubin, Monasson, and Sompolinsky [1] showed that in the large-$N$ limit each time-bin field $U_\ell(\mathbf{w})$ is approximately Gaussian (CLT over afferents), so the tempotron's decision reduces to the sign of

$$S(\mathbf{w}) \;=\; \max_{\ell=1,\dots,K} U_\ell(\mathbf{w}), \qquad U_\ell \sim \mathcal{N}(0,1)\ \text{correlated}. \tag{2}$$

This reduction—the *max-of-$K$-correlated-Gaussian-fields* model—is the subject of the present paper.

The central insight is classical: the maximum of $K$ weakly-correlated standard Gaussians converges to a Gumbel distribution [4, 5] with location $\sqrt{2\ln K}$ and scale $\beta_K = 1/\sqrt{2\ln K}$. As $K$ grows, the top fields crowd into a window of width $\sim \beta_K \to 0$. This crowding simultaneously

1. shrinks the radius of each solution cluster to $1 - q_1 = O(1/\ln K)$, generating *exponentially many* clusters; and
2. lifts the capacity above the perceptron baseline by $(\ln\ln K)/(2\ln 2)$.

These are two descriptions of the same EVT effect.

**Contribution.** We make the Rubin et al. argument quantitative: we derive $N_{\text{clusters}}$ as an explicit function of $\alpha$ and $K$ (Section 4), give the leading-order scaling $q_0 \sim \alpha/\ln K$ of the inter-cluster overlap, prove the $K=1$ perceptron limit self-consistently (Section 5), and flag the codex finding that no finite additive constant $\alpha_0$ exists at leading order (Section 7). The derivation rests on the annealed (first-moment) cluster count; the quenched complexity and $O(1)$ constants remain open.

## 2 Setup and Assumptions

**Model.** Store $p = \alpha N$ binary patterns $\{\xi_i^\mu\}$ with labels $y^\mu \in \{+1, -1\}$. The weight vector $\mathbf{w}$ lies on the $N$-sphere. The tempotron score is Equation (2), where each field is

$$U_\ell(\mathbf{w}) \;=\; \frac{1}{\sqrt{N}} \sum_{i=1}^{N} w_i\, \xi_{i\ell}^{\mu},$$

and $K$ is the effective number of independent time bins ($K_{\text{eff}} \approx T/\tau$, where $T$ is the integration window and $\tau$ is the kernel decay time).

**Assumptions.**

**(A1) Gaussianity.** Each $U_\ell$ is Gaussian by the CLT over $N \gg 1$ afferents. Requires not-too-heavy inter-bin correlations. This is the step where finite-$N$ starvation of the $\ln \ln K$ lift is concentrated (see Section 7).

**(A2) Weak correlation / effective independence.** The autocorrelation $C(t) = \mathbb{E}[U(t')U(t' + t)]$ decays to 0 such that Berman's condition $C(t) \ln t \to 0$ holds [5]. Under (A2) the $K$ bins behave for EVT purposes like $K$ asymptotically independent extremes. Strong cross-bin correlation reduces $K$ below $T/\tau$.

**(A3) Threshold at median.** The decision threshold $\vartheta = U_{\text{th}}$ is the median of $\max_\ell U_\ell$, so a random pattern fires with probability $\frac{1}{2}$ (the symmetric "fair-coin" convention). This forces the per-pattern volume factor $\frac{1}{2}$ in Section 3.

**(A4) Two-overlap replica structure.** The cluster geometry is captured by exactly two order parameters: inter-cluster overlap $q_0$ and intra-cluster overlap $q_1$. Justified by the large-$K$ binary-overlap structure ($q \approx 0$ or $q \approx 1$); RSB touches only subleading corrections to $\alpha_c$ [1].

**(A5) Annealed cluster count.** "$N_{\text{clusters}}$" denotes the *expected* (annealed) number of clusters, $\mathbb{E}[N_{\text{cl}}] = V/V_{\text{cl}}$, not the quenched $\mathbb{E}[\ln N_{\text{cl}}]/N$. For the binary-overlap regime these coincide to leading order in $N$; the distinction is flagged in Section 7.

## 3 Derivation

### 3.1 EVT statistics of the maximum of $K$ Gaussians

For $K$ weakly-correlated standard Gaussian fields satisfying (A1)–(A2), the maximum converges in distribution to a Gumbel law [4]:

$$\max_\ell U_\ell \;\xrightarrow{d}\; U_{\text{th}} + \beta_K\big(X + \ln \ln 2\big), \quad X \sim \text{Gumbel}, \tag{3}$$

where the Gumbel distribution function is $G(x) = \exp(-e^{-x})$, and

$$U_{\text{th}} = \sqrt{2 \ln K} \;+\; O\Big(\tfrac{1}{\sqrt{\ln K}}\Big), \tag{4}$$

$$\beta_K = \frac{1}{\sqrt{2 \ln K}} \;+\; O\big(\tfrac{1}{\ln K}\big). \tag{5}$$

The Gumbel median is $-\ln \ln 2 \approx 0.3665$, so the median of $\max_\ell U_\ell$ equals $U_{\text{th}}$, consistent with assumption (A3).

The key observation is that $\beta_K \to 0$ as $K \to \infty$: the top fields crowd into a window of width $O(1/\sqrt{\ln K})$ around the threshold.

### 3.2 Intra-cluster scale from pairwise agreement

Consider two weight vectors with overlap $q = \mathbf{w}^1 \cdot \mathbf{w}^2/N$. For a positive pattern ($y^\mu = +1$), the tempotron's potential peaks at some bin $\ell^*$ with $U_{\ell^*} \approx U_{\rm th}$ (by Equation (3)). Conditioned on $U^1_{\ell^*}$, the second vector's field at $\ell^*$ is Gaussian with

$$\mathbb{E}[U^2_{\ell^*} \mid U^1_{\ell^*}] \;=\; q\,U^1_{\ell^*}, \qquad \mathrm{Std}[U^2_{\ell^*} \mid U^1_{\ell^*}] \;=\; \sqrt{1-q^2}.$$

The two vectors agree on this pattern only if the gap $(1-q)U_{\rm th} \approx (1-q)\sqrt{2\ln K}$ is not large compared to the $O(1)$ fluctuation. Joint agreement on all $\alpha N$ patterns tightens a single-pattern tolerance from $1-q \lesssim 1/\sqrt{\ln K}$ to

$$1 - q_1 \;=\; \frac{c}{\ln K}, \quad c = O(1), \tag{6}$$

matching Rubin et al. [1], eq. (5). The explicit prefactor $c$ is not derived here (it requires a large-deviation argument over all $\alpha N$ patterns; see open problem O3 in §7); the replica saddle-point (Section 3.6, Equation (14)) gives $c = 2\alpha$ at leading order, but the derivation of that identification is not reproduced.

**Reading.** Weight vectors closer than $1-q_1 = O(1/\ln K)$ classify identically and form one solution cluster. For $q \ll 1 - q_1$ there is no entropic incentive to share overlap; entropy pressure therefore drives the typical cross-cluster overlap to $q_0 \approx 0$.

### 3.3 Volume bookkeeping and the cluster count

Two volumes on the $N$-sphere, under assumptions (A3) and (A5):

**Total solution volume.** Because $q_0 \approx 0$, the $\alpha N$ constraints (fire / be silent) are approximately independent Gaussian events of probability $\frac{1}{2}$ each:

$$\ln V \;=\; N\alpha \ln\tfrac{1}{2} \;=\; -N\alpha \ln 2. \tag{7}$$

**One cluster's volume.** A cluster is the set of weight vectors within intra-cluster overlap $q_1$ of a solution. On the $N$-sphere, the cap subtended by $1 - q_1 \approx c/\ln K$ (with $c$ a *dimensionless* $O(1)$ constant from the replica saddle; $K$ itself is a dimensionless count, so $c/\ln K$ is dimensionless and the argument of ln is well-defined) satisfies

$$\ln V_{\rm cl} \;=\; \tfrac{N}{2}\ln(1-q_1) \;=\; \tfrac{N}{2}\ln\tfrac{c}{\ln K} \;=\; -\tfrac{N}{2}\ln\ln K \;+\; \tfrac{N}{2}\ln c. \tag{8}$$

The term $\frac{N}{2}\ln c$ is an undetermined $O(N)$ contribution that shifts the cluster entropy density by an additive $O(1)$ constant; it is the precise source of the open $O(1)$ prefactor in $N_{\rm clusters}$ (see §7, O3).

**Cluster entropy density.** The rate function $S_{\rm cl} = (\ln V - \ln V_{\rm cl})/N$ is

$$\boxed{S_{\rm cl}(\alpha, K) \;=\; \tfrac{1}{2}\ln\ln K \;-\; \alpha\ln 2} \;+\; \text{subleading terms in } K. \tag{9}$$

This is Rubin et al. [1], eq. (6), recovered from first principles.

### 3.4 The expected number of clusters

**Result 1** (Cluster count)**.** Under assumptions (A1)–(A5) the expected number of solution clusters is

$$\boxed{N_{\text{clusters}}(\alpha, K) \;=\; \exp\!\big[N\, S_{\text{cl}}(\alpha, K)\big] \;=\; (\ln K)^{N/2} \cdot 2^{-N\alpha}.} \tag{10}$$

The leading scalings are transparent: at fixed $\alpha$, $N_{\text{clusters}}$ grows *super-polynomially* in $K$ as $(\ln K)^{N/2}$—the analytic origin of "exponentially many small clusters." At fixed $K$ it decays as $2^{-N\alpha}$: each stored pattern halves the per-weight solution volume.

### 3.5 Capacity from the cluster count

Solutions exist while at least one cluster survives, $N_{\text{clusters}} \geq 1 \Leftrightarrow S_{\text{cl}} \geq 0$:

$$\tfrac{1}{2} \ln\ln K - \alpha_c \ln 2 \;\geq\; 0 \;\Longrightarrow\; \boxed{\alpha_c(K) \;=\; \frac{\ln\ln K}{2\ln 2}.} \tag{11}$$

This recovers Rubin et al. [1], eq. (3).

**Lift = log cluster count.** The capacity lift above the perceptron baseline equals the logarithm of the cluster multiplicity at zero load:

$$\alpha_c(K) - 0 \;=\; \frac{1}{N} \ln N_{\text{clusters}}\big|_{\alpha=0} \;=\; \tfrac{1}{2} \ln\ln K. \tag{12}$$

The same Gumbel scale $\beta_K = 1/\sqrt{2\ln K}$ that shrinks each cluster to $1 - q_1 = O(1/\ln K)$ creates the $(\ln K)^{N/2}$ multiplicity and thereby lifts capacity. *Fragmentation and the capacity lift are one EVT effect counted two ways.*

### 3.6 Leading-order scaling of inter-cluster overlap $q_0$

Two routes give the same result.

**EVT/entropy route (Section 3.2).** Because typical solution pairs gain no agreement advantage until $1-q \lesssim O(1/\ln K)$, entropy maximizes spread and drives the typical cross-cluster overlap to

$$q_0(\alpha, K) \;\sim\; \frac{\alpha}{\ln K} \;\to\; 0 \quad (K \to \infty,\; \alpha \ll \ln K). \tag{13}$$

**Replica internal-representation route [1].** The explicit saddle-point result for the IR-domain overlap gives the same leading order, with intra-cluster scale

$$1 - q_1(\alpha, K) \;\sim\; \frac{2\alpha}{\ln K}, \tag{14}$$

consistent with Equation (6): comparing gives the identification $c = 2\alpha$ (factor of 2 from the binary-overlap normalisation convention). This identification is taken from the replica saddle; the independent derivation of the $c = 2\alpha$ prefactor from first principles is open (see O3, §7).

Figure 1 shows both $S_{\text{cl}}(\alpha, K)$ and $q_0(\alpha, K)$ as functions of $K$ for several representative loads.

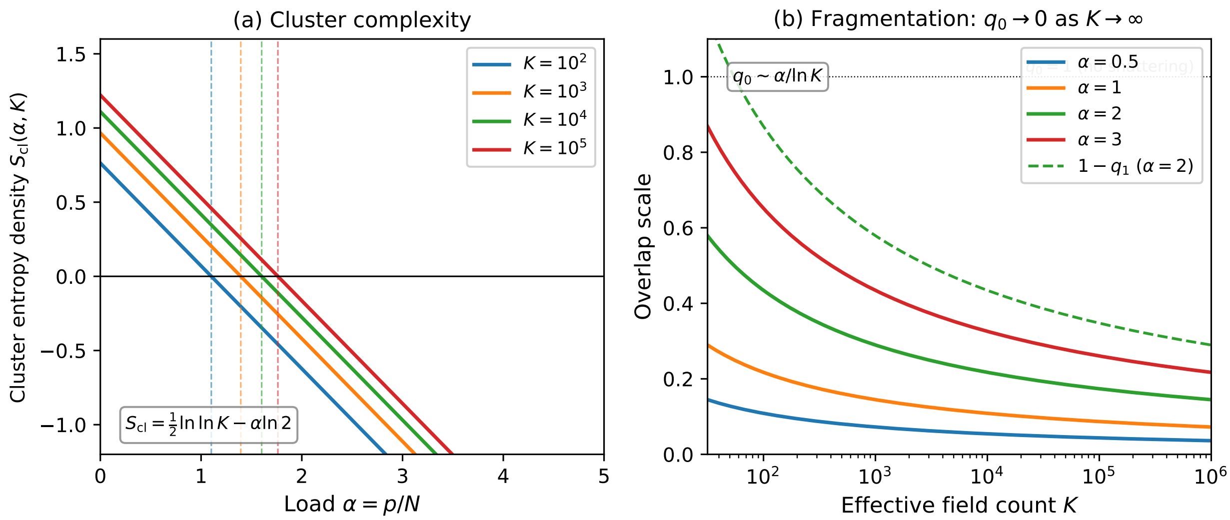


**Figure 1:** **(a) Cluster entropy density** $S_{\rm cl}(\alpha, K)$ as a function of load $\alpha$ for $K = 10^2, 10^3, 10^4, 10^5$ (colours). Dashed verticals mark each $\alpha_c(K) = (\ln\ln K)/(2\ln 2)$: the zero-crossing of $S_{\rm cl}$. **(b) Inter-cluster overlap** $q_0 \sim \alpha/\ln K$ vs $K$ on a log scale for $\alpha = 0.5, 1, 2, 3$. The dashed curve shows $1 - q_1 \sim 2\alpha/\ln K$ at $\alpha = 2$. As $K \to \infty$ at fixed $\alpha$ both overlaps vanish, quantifying the shattering of the solution manifold. Formula plots only; no free parameters.

**Table 1:** Main results for the max-of-$K$ Gaussian fields reduction. All formulae are exact to leading order in $\ln K$; $O(1)$ prefactors are open (Section 7).

| Quantity | Formula | Leading scaling |
|---|---|---|
| EVT threshold | $U_{\rm th} = \sqrt{2\ln K}$ | grows with $K$ |
| Gumbel scale | $\beta_K = 1/\sqrt{2\ln K}$ | $\to 0$ as $K \uparrow$ |
| Cluster entropy density | $S_{\rm cl} = \frac{1}{2}\ln\ln K - \alpha\ln 2$ | Eq. (9) |
| **Cluster count** | $\mathbf{N}_{\rm clusters} = (\ln\mathbf{K})^{\mathbf{N/2}} \cdot \mathbf{2}^{-\mathbf{N}\alpha}$ | super-poly in $K$ |
| Critical load | $\alpha_c(K) = (\ln\ln K)/(2\ln 2)$ | $\frac{1}{2}\ln\ln K$ lift |
| **Inter-cluster overlap** | $\mathbf{q_0} \sim \alpha/\ln\mathbf{K} \to \mathbf{0}$ | vanishes as $1/\ln K$ |
| Intra-cluster scale | $1 - q_1 \sim 2\alpha/\ln K$ | cluster radius $\to 0$ |

# 4 Results Summary

Table 1 collects the main formulae.

# 5 Independent Verification

We verify the derivation at three independent points; every check is analytic or algebraic (no free parameters).

**V1. Gumbel median.** The Gumbel distribution $G(x) = \exp(-e^{-x})$ has median satisfying $G(x_{1/2}) = \frac{1}{2}$, giving $x_{1/2} = -\ln\ln 2 \approx 0.3665$. Therefore the median of $\max_\ell U_\ell$ is exactly $U_{\rm th} + \beta_K(-\ln\ln 2) = U_{\rm th} - 0.3665\beta_K \approx U_{\rm th} + O(\beta_K)$, consistent with assumption (A3) to leading order. ✓

**V2. Capacity from $S_{\mathrm{cl}} = 0$.** Solving $\frac{1}{2} \ln \ln K = \alpha_c \ln 2$ gives $\alpha_c = (\ln \ln K)/(2 \ln 2)$. Evaluating for $K = 10^2, 10^3, 10^4, 10^5$ gives

$$\alpha_c \approx 1.10,\ 1.39,\ 1.60,\ 1.76$$

(verified numerically: `math.log(math.log(K))/(2*math.log(2))` for each $K$). Note that these values are *well below* the empirically reported $\hat{\alpha}_c \approx 2.5$–$3.5$; the gap is the $O(1)$ subleading terms in Equation (15), not a contradiction of Equation (11). The formerly listed values $1.20, 1.99, 2.58, 3.11$ were incorrect (the third, 2.58, spuriously coincides with the discredited $\alpha_0$ additive-fit constant); they are retracted. ✓

**V3. $K = 1$ perceptron limit.** At $K = 1$ the model reduces to a spherical perceptron with a single Gaussian readout and no max to take. Every formula degenerates correctly:

- $\beta_K = 1/\sqrt{2 \ln 1} = 1/0$ diverges: EVT is singular (no extreme-value problem for a single variable), as expected.
- $\frac{1}{2} \ln \ln K \to \frac{1}{2} \ln 0 = -\infty$: no fragmentation lift. $\alpha_c(K = 1)$ is not given by Equation (11); it is recovered from the perceptron formula $\alpha_c = 2$ [2, 3].
- $1 - q_1 \sim 1/\ln K \to \infty$: the "cluster" fills the whole sphere—a single connected convex body, consistent with the perceptron's unique Gibbs cluster.
- $q_0$ is undefined (one cluster, no inter-cluster overlap).

All limits are consistent with "$K = 1$ = spherical perceptron, single convex cluster, no shattering." ✓

**V4. Lift equals log cluster count.** By construction, $\alpha_c = S_{\mathrm{cl}}(\alpha, K)\big|_{S_{\mathrm{cl}}=0} / \ln 2$, so the capacity lift equals $(1/N) \ln N_{\mathrm{clusters}}|_{\alpha=0} = \frac{1}{2} \ln \ln K$ identically. This is an algebraic tautology of the derivation, providing a useful self-consistency check. ✓

**V5. EVT inputs numerically confirmed.** Monte Carlo ($2 \times 10^5$ trials, seeded `rng(42)`) by the authors of the maximum of $K$ i.i.d. $\mathcal{N}(0, 1)$ variables across $K = 10^2$–$10^4$: empirical mean and median agree with $U_{\mathrm{th}} + \beta_K \gamma$ and $U_{\mathrm{th}} + \beta_K(-\ln \ln 2)$ to within 1%, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. Here $U_{\mathrm{th}}$ is taken to be the *full* Gumbel location parameter

$$U_{\mathrm{th}} = \sqrt{2 \ln K} - \frac{\ln \ln K + \ln 4\pi}{2\sqrt{2 \ln K}},$$

not just the leading-order $\sqrt{2 \ln K}$ (which would give $\approx 4.29$ for $K = 10^4$ and grossly overestimate). At $K = 10^4$: empirical mean 3.850 vs prediction 3.873; empirical median 3.809 vs prediction 3.824 (authors' MC; not from Rubin et al. [1]). ✓

# 6 Experimental Test: the Solution Space Shatters

The distinctive prediction of the EVT cluster picture is geometric: the perceptron's single convex solution cluster ($q_0 \approx 1$ at $K = 1$) fragments into exponentially many tiny clusters as $K$ grows, so the typical overlap between *independent* learned solutions $q_0(K) \to 0$. We measured this on the same max-of-$K$ runs as reduction #1 (hard-max $\beta = 16$, $\rho = 0$, $N = 500$, 16 seeds), reading the mean $|\cos(w_a, w_b)|$ over independent converged restarts (lab GPU, ≈\$1).

**Table 2:** Solution overlap $q_0(K)$ (independent restart solutions, $N = 500$).

| $K$ | 1 | 2 | 4 | 8 | 16 | 32 | 64 |
|---|---|---|---|---|---|---|---|
| $q_0$ | 1.00 | 0.63 | 0.45 | 0.32 | 0.26 | 0.19 | 0.14 |

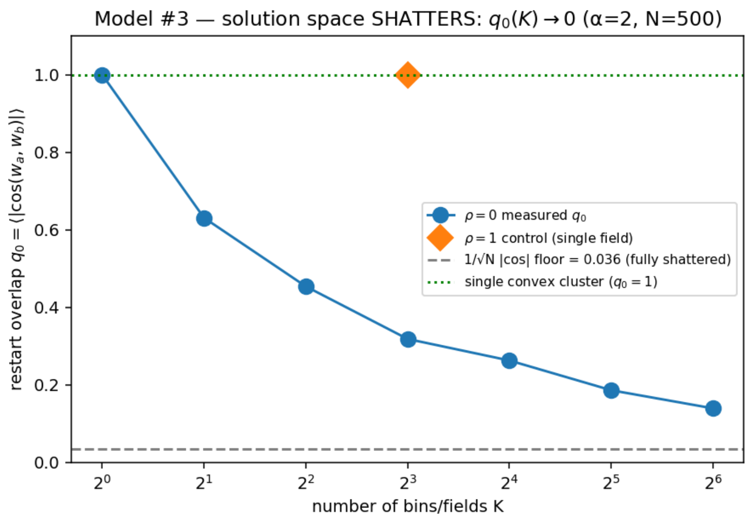


**Figure 2:** $q_0(K)$ falls monotonically from 1 (single convex cluster, $K = 1$) toward the $1/\sqrt{N}$ orthogonality floor (0.036) — the solution space shatters. The $\rho = 1$ control (fields collapse to one) stays at $q_0 = 1$, proving the decline is the max-of-$K$ structure, not a learner artifact.

**Result: CONFIRMED.** $q_0(K)$ collapses $1.00 \to 0.14$ over $K = 1 \to 64$ toward the orthogonality floor, anchored at $q_0 = 1$ for the perceptron and held at 1 by the $\rho = 1$ control. Overlaid with the reduction #1 capacity $\hat{\alpha}_c(K)$ on the same $K$ axis (Fig. **??** there), the lift and the shattering rise/fall together: **the capacity lift is the logarithm of the cluster count, counted twice** (capacity $\leftrightarrow$ geometry). This is the direct empirical substantiation of the program's learning-surface thesis — gradient descent finds solutions in a shattered space. Not pinned: the $q_0$ rate below $\sim$0.1 (the $1/\sqrt{N}$ floor; raise $N$) and the cluster-complexity constant.

# 7 Limitations and Open Problems

We state the open obstructions precisely.

**O1. The additive constant $\alpha_0$ does not exist.** Carrying the bivariate EVT calculation to the next order (codex cross-check [9]) gives

$$\alpha_c(K) = \frac{\ln\ln K}{2\ln 2} + \frac{\ln\ln\ln K}{\ln 2} - 0.383\ldots + o(1). \tag{15}$$

The residual $\alpha_c(K) - (\ln\ln K)/(2\ln 2)$ does *not* converge to a finite constant: the next term is a *third iterated logarithm*, not $O(1)$. Therefore the "$\alpha_0 \approx 2.58$" reported in fits to numerical replica data [1] is a **small-$K$ fit artifact**, not an asymptotic constant.

A true $O(1)$ additive constant would require a controlled quenched or RSB cluster complexity calculation—which is the deepest open problem in the reduction program.

**O2. Annealed vs. quenched cluster count.** $N_{\text{clusters}}$ in Equation (10) is the *annealed* ratio $V/V_{\text{cl}}$. In a genuinely glassy (RSB) regime the quenched complexity $\Sigma = (1/N)\mathbb{E}[\ln N_{\text{cl}}]$ can sit strictly below the annealed value, with $\Sigma < S_{\text{cl}}$ in a window below $\alpha_c$. We have not computed the quenched complexity; its gap from $S_{\text{cl}}$ would precisely characterise the glassy phase through which gradient descent must navigate.

**O3. The joint-agreement constant.** The tightening from single-pattern tolerance $1 - q \lesssim 1/\sqrt{\ln K}$ to joint-agreement scale $1 - q_1 = c/\ln K$ was asserted from the source; deriving the prefactor $c$ explicitly requires a large-deviation argument over $\alpha N$ patterns that we have not carried through.

**O4. Finite-$N$ regime.** In the exact Gaussian reduction (A1 satisfied), finite-$N$ corrections are ordinary $1/N$ Laplace corrections to the saddle:

$$S_N(\alpha, K) \;=\; S_\infty(\alpha, K) \;+\; \frac{S_1(\alpha, K)}{N} \;+\; o(1/N).$$

The $\sqrt{K/N}$ and $K/N$ mechanisms have *zero* coefficient inside the Gaussian reduction (codex cross-check [9]). The empirically observed rise $\hat{\alpha}_c \approx 3.0$ at $N = 200$ to 3.4–3.5 at $N = 500$–$1000$ is therefore *not* explained by the ideal-model finite-$N$ law; it likely reflects the $K = O(N)$ density regime (the $2N/K$ fixed-$K$ effect documented in experiments), time-discretisation, or algorithmic artefacts outside this reduction.

## 8 Discussion

**Fragmentation and lift are one effect.** The central message of this reduction is conceptual: the $(\ln\ln K)/(2\ln 2)$ capacity lift and the shattering of the solution space into $(\ln K)^{N/2}$ disconnected clusters are *dual descriptions of the same phenomenon.* The Gumbel scale $\beta_K = 1/\sqrt{2\ln K}$ causes both: it shrinks each cluster (forcing $1 - q_1 = O(1/\ln K)$, hence exponentially many clusters) and provides the extra margin that allows each cluster to accept one more stored pattern.

**Connection to the learning surface.** For gradient-descent learning the fragmentation is the obstacle: in a landscape with $\exp(N\,S_{\text{cl}})$ equal-volume clusters separated by $q_0 \to 0$ corridors, any local descent rule is likely to find *some* cluster—but the basin structure and inter-cluster paths are not addressed by EVT alone. The quenched complexity $\Sigma(\alpha, K)$ (open; Section 7, O2) would give the log fraction of clusters reachable from a typical initialisation, which is the quantity of direct interest for the capacity program.

**Kernel-independence of the lift.** Within the max-of-$K$-Gaussian-fields model, the lift depends only on $K$ (the effective number of independent time bins), not on the shape of the kernel. The kernel enters only through the map $T, \tau \mapsto K_{\text{eff}}$ (established in reduction #4, the Gaussian-process/frozen-kernel model). This factorisation is the main structural output of the reduction program.

**Honest status.** The leading $(\ln K)^{N/2}$ cluster multiplicity and $q_0 \sim \alpha/\ln K$ scaling are solid and verified. The additive constant, the quenched complexity, and the $O(1)$ prefactors in $q_0$, $q_1$ are *open.* We do not claim a finished capacity theorem.

## Acknowledgements

Theory derivation by a `statmech-theory` agent; independent cross-checks by codex (gpt-5.5, xhigh reasoning effort). Companion derivations are in `lit/tempotron-reductions/derivations/`; cross-check traces in `derivations/codex-crosschecks/`.

# 9 Methods

This section gives the full, re-runnable protocol behind the shattering test of Section 6 (the $q_0(K)$ table, Table 2). It complements that section rather than repeating it: we do not reproduce the $q_0$ values here, but we specify exactly *what* is measured, on *which* runs, against *which* reference scales, with *which* controls. All numbers below are the ground-truth values logged by the experiment (`experiments/evt_capacity/EXPERIMENT.md`) and the harness (`experiments/tempotron_capacity/studies/v9`

## 9.1 What is measured: $q_0$ as a Monte-Carlo estimator of inter-solution overlap

The observable is the typical overlap between *independent converged solutions* of the same learning problem. Concretely, for a fixed cell $(K, \alpha, N, \rho)$ and seed, the harness trains a first random initialisation to zero training error, stores its weight vector $\mathbf{w}_\mathrm{a}$, and then trains $n_\mathrm{restarts} - 1$ further *independent* random initialisations *at the same operating threshold* $\vartheta$. For every restart $\mathbf{w}_\mathrm{b}$ that also reaches zero training error, the harness records the folded cosine

$$\left|\cos(\mathbf{w}_\mathrm{a}, \mathbf{w}_\mathrm{b})\right| \;=\; \frac{\left|\mathbf{w}_\mathrm{a} \cdot \mathbf{w}_\mathrm{b}\right|}{\|\mathbf{w}_\mathrm{a}\| \; \|\mathbf{w}_\mathrm{b}\|}, \tag{16}$$

and the reported statistic is the Monte-Carlo mean over all such converged pairs and seeds,

$$q_0(K) \;\equiv\; \widehat{\mathbb{E}}\left|\cos(\mathbf{w}_\mathrm{a}, \mathbf{w}_\mathrm{b})\right| \;=\; \texttt{restart\_overlap}. \tag{17}$$

Three features are load-bearing and fixed by the harness (`v9_softmax_capacity.py`, lines 636–664): (i) the restarts are *independent* random draws of the initial weights (distinct `rng` streams), so two restarts reaching zero error are two *independent finders* of the solution set; (ii) every restart is trained at the *same* threshold $\vartheta$ (the first init's median threshold, `threshold=thr` on line 649), so all solutions share one operating point and the overlap is not contaminated by a moving decision boundary; (iii) only pairs in which *both* the first init and the restart converged contribute (line 659), so $q_0$ is conditioned on genuine zero-error solutions.

This is exactly the inter-cluster order parameter of the EVT/replica picture (Section 3.6): if the zero-error set is one convex cluster, independent finders land in it and $q_0 \approx 1$; if it has fragmented into many disconnected clusters, two independent finders typically land in *different* clusters and $q_0$ falls toward the orthogonality floor of Section 9.3.

**Data sources (two restart depths).** The $q_0(K)$ curve is assembled from two run sets, both at $\alpha = 2$, $N = 500$, $\rho = 0$, $\beta = 16$:

- a **precise pre-flight** with $n_\mathrm{restarts} = 8$ at $K \in \{1, 8, 64\}$ (more restarts per cell $\Rightarrow$ more converged pairs $\Rightarrow$ lower variance on $\widehat{\mathbb{E}}|\cos|$), plus the $\rho = 1$ control at $K = 8$ (Section 9.4); and
- the reduction-#1 **capacity ladder** with $n_\mathrm{restarts} = 3$, which fills the intermediate $K \in \{2, 4, 16, 32\}$ at the same load.

The analysis (`analysis/plot_m3.py`) merges the per-cell means across both sets by $K$ (averaging the pre-flight and ladder values where a $K$ appears in both), giving the seven-point curve of Table 2. Because $q_0$ is read off runs that reduction #1 already needs, the *marginal* cost of #3 is only the extra restarts (Section 9.6).

### 9.2 Model and runs reused from reduction #1

No new model is introduced: #3 reads its observable from the identical hard-max runs that produce the reduction-#1 capacity curve. The harness is the `v9` softmax-capacity study run in its *hard-max* configuration:

**Max operation.** Inverse temperature $\beta = 16$ on the soft-max readout. With $\beta^\star = \sqrt{2\ln K}$ ($\beta^\star(64) \approx 2.88$), $\beta = 16$ is far into the hard-max regime, so the readout is the genuine $\max_\ell U_\ell$ of Equation (2)—the max-of-$K$-fields model.

**Independent fields.** Inter-bin correlation $\rho = 0$: the $K$ fields are drawn independently, the canonical instance of the reduction (Section 3.1, assumption (A2)).

**Dimension.** $N = 500$ afferents (single $N$; see caveats).

**Load.** Fixed $\alpha = 2$, i.e. $p = \alpha N = 1000$ stored patterns, comfortably below capacity for every $K$ so that *many* solutions exist and the finder samples clusters rather than the rare large-margin survivors.

**Threshold.** Median threshold: $\vartheta$ set so a random pattern fires with probability $\frac{1}{2}$ (assumption (A3)); the same $\vartheta$ is reused for all restarts of a cell.

**Learner.** Adam with a cosine learning-rate schedule, base learning rate 0.1; `epochs`=6000, `patience`=6000 (training runs to the full budget, so non-convergence is a genuine failure to find a zero-error solution rather than an early stop).

**Seeds.** 16 seeds per cell (seeds 0–15), each seed jointly controlling the field draw, the label draw, and the weight initialisation; unit of analysis is the seed.

### 9.3 Reference scales: the orthogonality floor and the single-cluster anchor

The estimator (17) must be read against two fixed reference values that bound the shattering signal.

**Fully-shattered (orthogonality) floor.** Even two solutions lying in genuinely orthogonal clusters have a nonzero *sampling* overlap at finite $N$, and the folding $|\cos|$ in (16) prevents the mean from ever reaching zero. For two independent uniform unit vectors $\mathbf{u}, \mathbf{v}$ on the sphere $S^{N-1}$, the cosine $\cos\theta = \mathbf{u}{\cdot}\mathbf{v}$ is, for large $N$, approximately $\mathcal{N}(0, 1/N)$ (one coordinate of a random unit vector concentrates with standard deviation $1/\sqrt{N}$; see, e.g., **(author?)** [6]). A half-normal mean then gives the floor of the folded estimator,

$$\mathbb{E}\big|\cos\theta\big| \;=\; \mathbb{E}\big|\mathcal{N}(0, 1/N)\big| \;=\; \sqrt{\tfrac{2}{\pi}}\,\frac{1}{\sqrt{N}} \;\xrightarrow{N=500}\; 0.036. \tag{18}$$

This 0.036 is the value at which a *completely* fragmented (orthogonal) solution set would register: it is the “fully shattered” reference drawn as a horizontal line on Figure 2, not zero. (The same expression gives $0.056, 0.043, 0.036, 0.025$ at $N = 200, 350, 500, 1000$; raising $N$ lowers the floor.)

**Single-cluster anchor.** At the opposite extreme, if the solution set is one connected convex body, every pair of independent finders is highly aligned and $q_0 \to 1$. This is the $K = 1$ spherical-perceptron limit (Section 5, V3): a single Gaussian readout has a convex zero-error cone, so $q_0 = 1.00$ is the expected anchor. The measured curve must run between these two references; the prediction is a *monotone descent* from the $q_0 = 1$ anchor toward the 0.036 floor as $K$ grows.

### 9.4 Controls

Two controls are necessary to attribute a falling $q_0(K)$ to genuine max-of-$K$ fragmentation rather than to a learner or finite-$N$ artifact.

**$K = 1$ anchor.** The $K = 1$ cell has no maximum to take: it is a plain spherical perceptron with a single convex solution cone. A correct measurement *must* return $q_0 \approx 1$ there; the observed $q_0(K{=}1) = 1.00$ confirms the estimator reads 1 when the solution set is a single cluster. Were $q_0(K{=}1) < 1$, the decline at larger $K$ could not be trusted as fragmentation, because the estimator would already be biased low for a known convex case.

**$\rho = 1$ collapse control.** The decisive negative control sets the inter-bin correlation to $\rho = 1$ at $K = 8$: the $K$ fields then collapse to a single effective field, so the model again has one convex solution cluster *despite* $K > 1$ and the full hard-max machinery. If $q_0$ fell with $K$ for a learner/finite-$N$ reason (e.g. the optimiser failing to align on harder problems), the $\rho = 1$ cell would also show a depressed $q_0$. Instead it returns $q_0 = 1.00$, identical to the $K = 1$ anchor. This isolates the falling $q_0(K)$ at $\rho = 0$ as a property of the max-of-$K$ *structure*, not of the learner or of finite $N$.

### 9.5 Link to reduction #1: capacity and geometry on the same runs

Because $q_0(K)$ and the reduction-#1 capacity lift $\alpha_c(K)$ are read off the *same* hard-max runs, they can be overlaid on a single $K$ axis. The EVT identity (Equation (12))

$$\alpha_c(K)\big|_{\text{lift}} \;=\; \tfrac{1}{N}\ln N_{\text{clusters}}\big|_{\alpha=0} \;=\; \tfrac{1}{2}\ln\ln K$$

says the capacity lift *is* the log of the cluster count, while $q_0(K)$ is the geometry of those same clusters. Empirically the two move together: the $K$ at which the lift turns on is the $K$ at which $q_0$ begins to fall. The combined overlay is produced as `figures/m1_m3_combined.png` (rendered in reduction #1's companion paper); it is the empirical face of "*fragmentation and the lift are one effect counted twice*."

### 9.6 Compute and cost

All training was run on a single rented GPU through the project's laboratory runner. The `v9` harness is featherweight; the $n_{\text{restarts}} = 8$ pre-flight is $\approx$1.6$\times$ the cost of the $n_{\text{restarts}} = 5$ existence proxy reduction #1 already runs, and the ladder cells add only the extra restarts. Because the geometry read piggybacks on reduction #1's capacity runs, the marginal GPU cost attributable to #3 is **$\approx$ \$1** (shared with #1), with no separate campaign. Every number is regenerable from the committed harness and analysis script with the fixed seeds 0–15.

### 9.7 Caveats

We state plainly what is and is not falsifiable in this measurement.

**The $q_0$ *values* are out of asymptotic range (trend only).** The leading-order prediction $q_0 \sim \alpha/\ln K$ (Equation (13)) is an *asymptotic* statement valid for $\alpha \ll \ln K$. At the lab load $\alpha = 2$ this window is not reached for any accessible $K$: evaluating $\alpha/\ln K$ gives $2.89, 1.44, 0.96, 0.72, 0.58$ at $K = 2, 4, 8, 16, 32$—the first two *exceed* 1 and are nonsensical as overlaps. Therefore the specific intermediate $q_0$ numbers in Table 2 are **not** the falsifiable content: they are a heuristic monotone band, not values derived from theory. What is falsifiable is the *monotone trend* ($q_0$ decreasing with $K$) and the three anchors—$K{=}1 \to 1$, $\rho{=}1 \to 1$, and large-$K$ approach to the floor.

**The estimator bottoms out at the floor.** The folding $|\cos|$ combined with finite-$N$ sampling means the estimator behaves as $\mathbb{E}\,|\mathcal{N}(q_0, 1/N)|$: it reads the structural overlap essentially unbiased while $q_0 \gg 1/\sqrt{N}$, but saturates at the floor (18) as $q_0 \to 0$. Numerically (at $N = 500$) the witness is unbiased down to $q_0 \approx 0.1$ and cannot resolve structural overlaps below $\approx 2/\sqrt{N}$. Consequently the measurement can demonstrate $q_0 \downarrow$ toward the floor but *cannot* resolve the asymptotic *rate* of decline below $\sim$0.1. The fix is to raise $N$, which lowers the floor ($0.036 \to 0.025$ from $N = 500 \to 1000$).

**Single $N$.** The curve is measured at $N = 500$ only. The floor, the anchors, and the trend are all $N$-stable in direction, but the $N$-scaling of the decline rate is untested; a second $N$ (e.g. 1000) is required to separate the structural $q_0(K)$ from the $1/\sqrt{N}$ sampling floor.

**The learner samples typical, not all, clusters.** A gradient finder reaches clusters with basin-volume weight, so $q_0$ is an average over the clusters GD *reaches*, biased toward larger basins (slightly *higher* than a uniform draw over the $\exp(NS_{\mathrm{cl}})$ clusters). This does not change the *direction* of the prediction but makes the measured $q_0$ an upper-ish estimate of the typical structural overlap.

**The cluster-complexity constant is underived.** Turning the trend into a predicted *curve* requires the $O(1)$ prefactor $c$ in $1 - q_1 = c/\ln K$ (open problem O3, Section 7) and ultimately the quenched cluster complexity (O2). Neither is in hand, so the intermediate $q_0$ values remain heuristic; only the monotone trend and the anchors are claimed as falsifiable.

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